

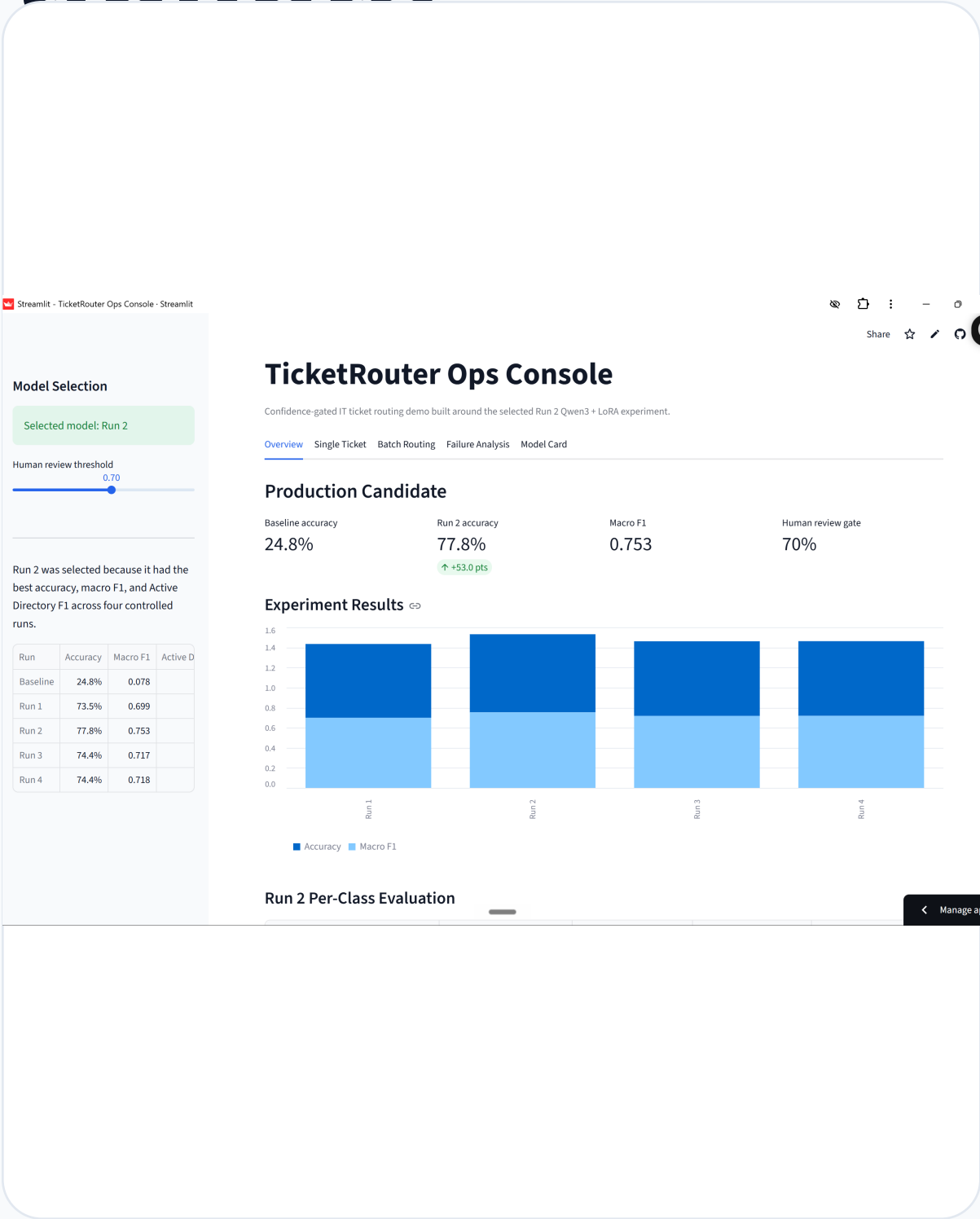
I fine-tuned a 1.7B model to route IT support tickets

TicketRouter turns messy helpdesk tickets into resolver-queue decisions using Qwen3-1.7B + LoRA.

24.8% baseline

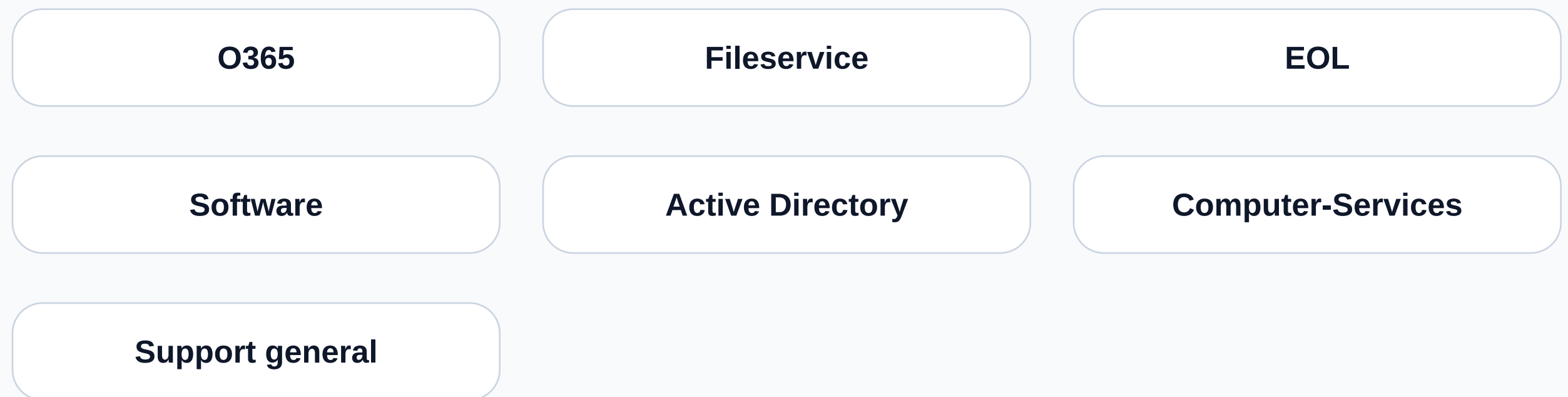
77.8% fine-tuned

+53.0 percentage points



The problem was not generating answers. It was routing work.

Internal IT tickets arrive through email, Slack, and portals. Before anyone can solve them, someone has to classify the request.



The model's job: first-pass triage only.

No customer replies. No final approvals. Just route the ticket to the right queue.

I chose LoRA because routing is behavior, not lookup.

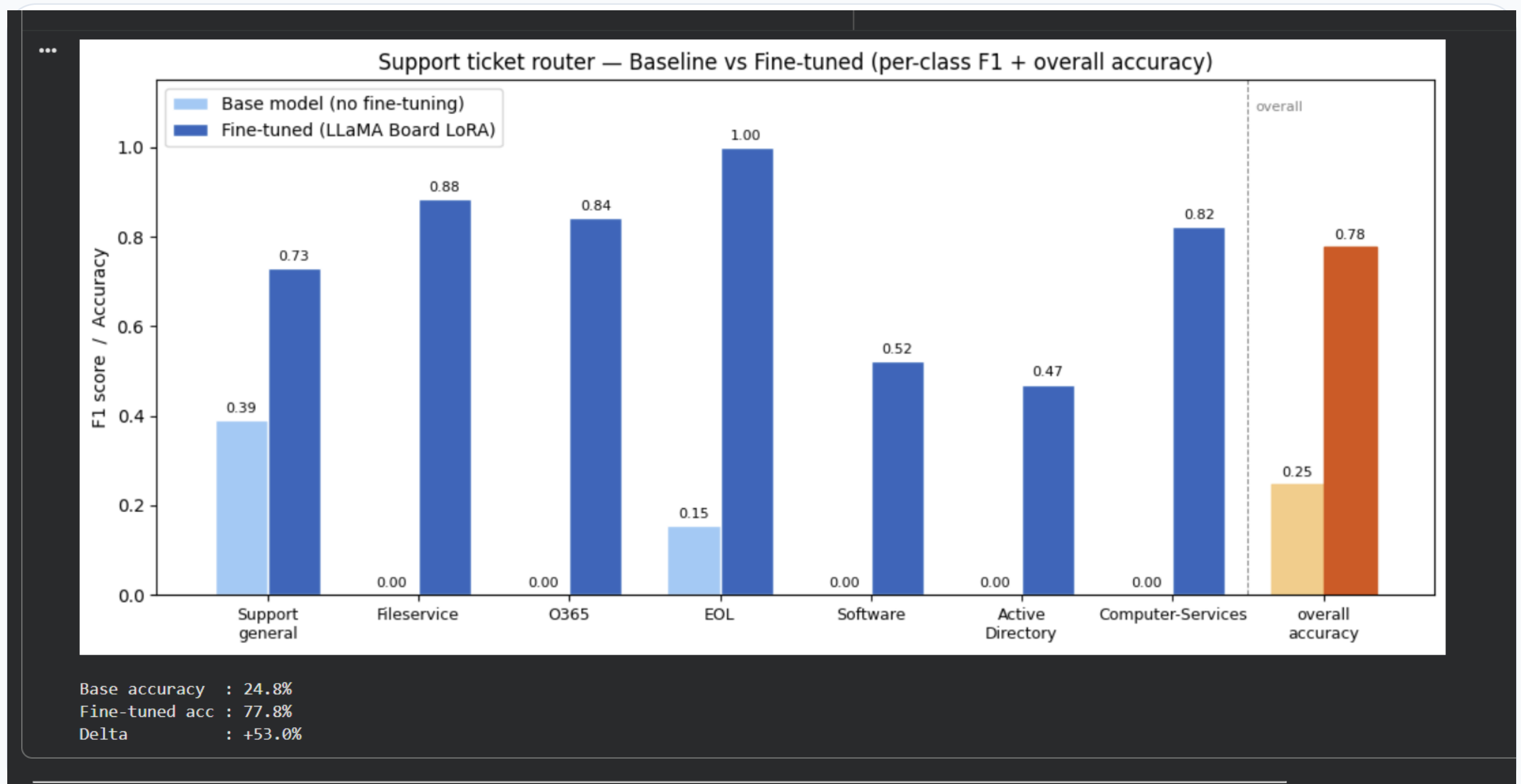
RAG was not the right fit. The system needed consistent classification patterns, not retrieved facts.



Result: lower training cost, smaller artifact, and enough capacity to learn IT routing language.

Fine-tuning created routing capability the base model did not have.

Run 2 won



24.8% -> 77.8% accuracy

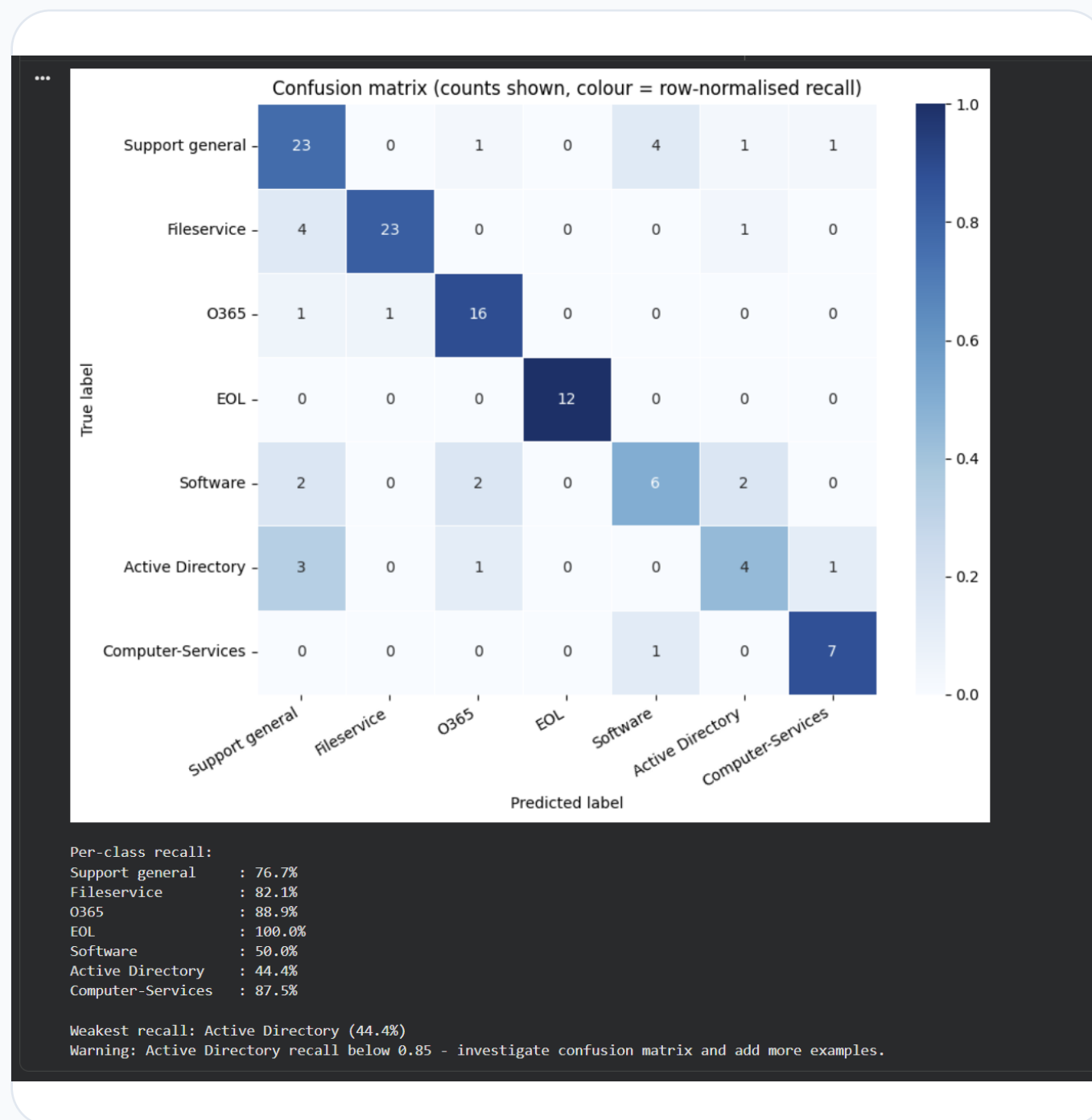
A +53.0 point lift over the base model, with macro F1 reaching 0.753.

Four runs turned this into an engineering decision.

Run	Accuracy	Macro F1	Verdict
Run 1	73.5%	0.699	Baseline established
Run 2	77.8%	0.753	Production candidate
Run 3	74.4%	0.717	Boundary data regressed
Run 4	74.4%	0.718	Epoch hypothesis rejected

The best story was not just the score. It was proving that more examples and more epochs did not automatically improve the model.

The failure analysis found the real bottleneck.



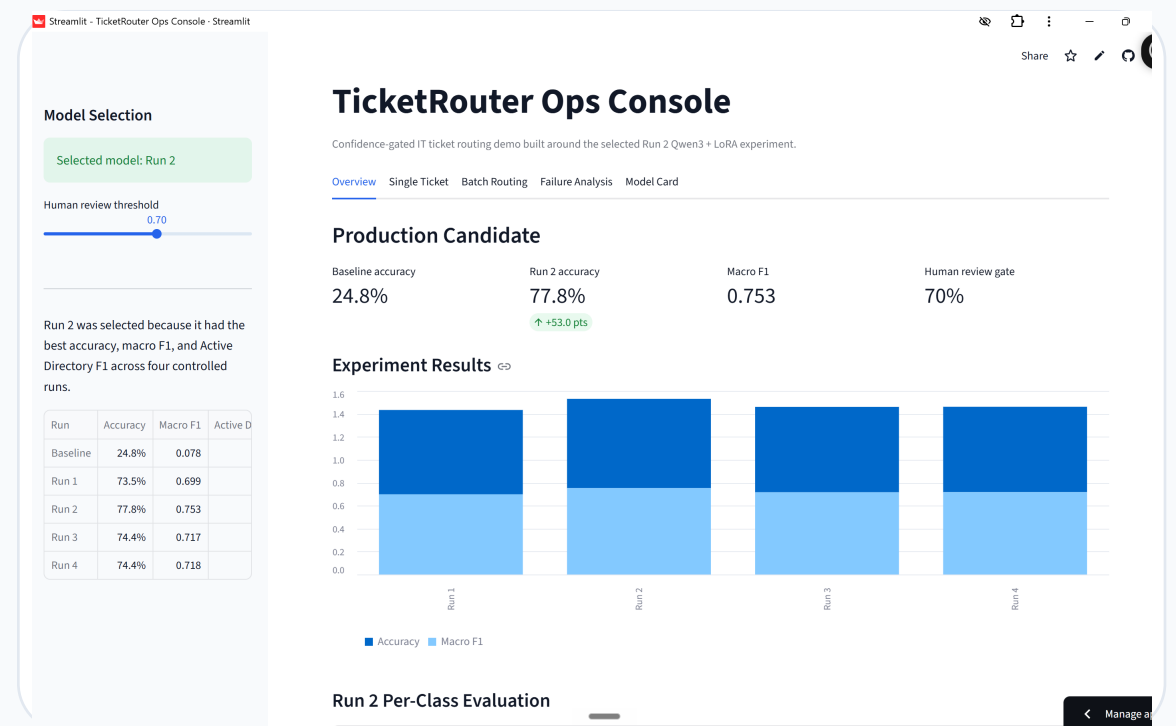
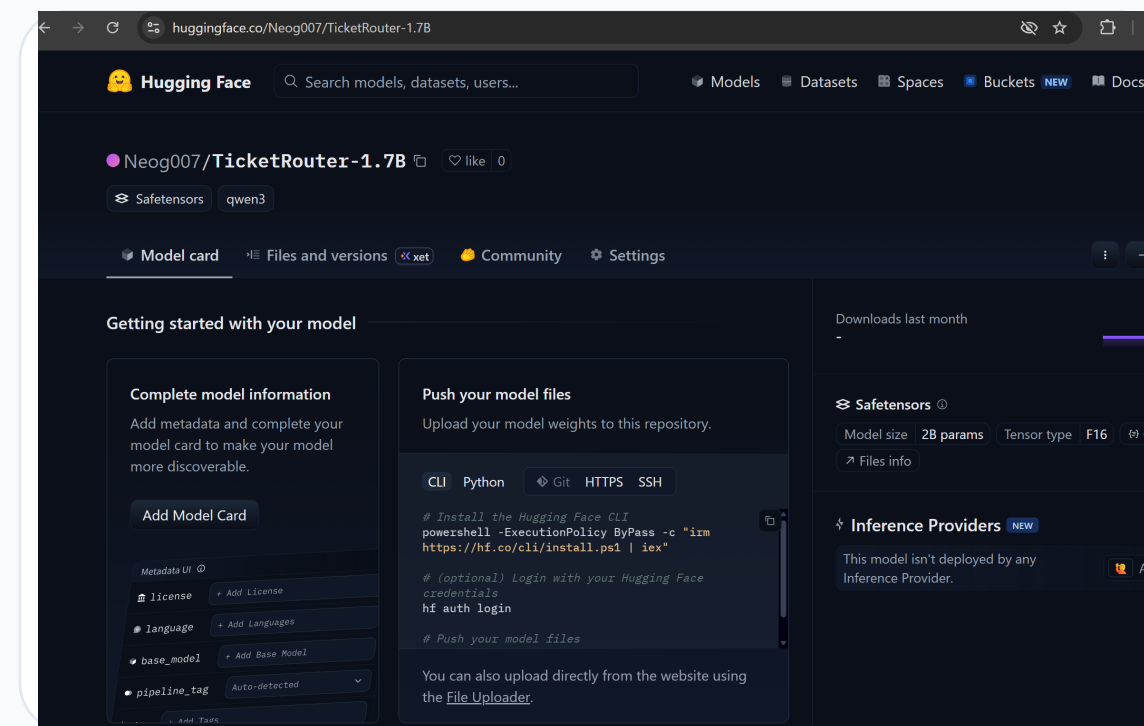
Weakest class:
Active Directory

Example miss: a contractor finance-group access request was predicted as O365 instead of Active Directory.

The fix is better AD and Software examples, not simply better training data.

Data quality > data volume

The final deliverable was a model decision, not just a notebook.



- Merged model artifact hosted on Hugging Face
- Streamlit ops-console demo for the workflow
- Model card with limitations and rollback plan
- Controlled experiment narrative for why Run 2 won

Builder lesson: trust evidence, not vibes.